

INSTITUTIONAL DETERMINANTS OF PRIVACY DESIGN IN CENTRAL BANK DIGITAL CURRENCIES

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Abstract

Central bank digital currencies (CBDCs) make transaction privacy an explicit institutional design choice. We ask whether political and economic conditions predict the privacy posture documented by central banks, while separating privacy vocabulary from design commitments. A frozen multilingual corpus contains 113 official or authority-attributed documents (3,963 pages) across forty-seven jurisdictions and two institutional composites. Two independently run large-language-model extractors produced 11,358 source-verified statements, deduplicated into 6,949 candidates. We aggregate them with a predeclared equal-vote ensemble in which every candidate has unit mass and provider disagreements are retained rather than averaged away. On a separate calibration set, the verified union attained precision 0.980, recall 0.766 and span fidelity 1.000; it failed the preregistered recall gate and is therefore treated as exploratory, not automated gold. In the production union, the privacy-family share and documented privacy posture correlate only $r = 0.219$. No tested institutional condition survives family-wise correction; the largest commitment-side association is with the informal economy ($r = -0.286$). A pre-frozen 5.25% double-blind human audit remains to be inserted before submission. The design demonstrates both the value and the limits of reproducible multi-model qualitative extraction.

Key words: central bank digital currency, privacy, content analysis, large language models, reproducibility

JEL Code: E42, E58, C83

Introduction

Around nine in ten central banks have explored a CBDC, while a smaller group has moved into pilots or live deployment (Kosse & Mattei, 2022). As the question shifts from feasibility to design, privacy becomes consequential. Physical cash normally settles without creating a central record of who paid whom. A retail CBDC may reproduce part of that boundary or make every transaction visible to intermediaries and public authorities. The outcome is a policy choice, not a technical inevitability.

Payment privacy has welfare content. Cash protects buyers from routine identification (Kahn, McAndrews, & Roberds, 2005), and privacy can have public-good properties that private payment markets underprovide (Garratt & van Oordt, 2021). CBDC design nevertheless combines this objective with financial integrity, monetary control and operational resilience. Agur, Ari, and Dell’Ariccia (2022) formalise the trade-off between cash-like and deposit-like features; Auer and Böhme (2020) show how architecture and access shape data exposure. Cross-country differences should therefore be observable in official design documents.

The empirical difficulty is measurement. A document may repeatedly promise privacy while preserving full state visibility. Counting privacy words then measures framing, not informational exposure. The same text can also discuss alternatives, cite external research or report a pilot without adopting the stated feature. Large multilingual corpora intensify these problems: manual census coding is costly, but unconstrained language-model output can miss eligible passages, invent unsupported quotations or confuse proposals with decisions. Recent research shows that large language models can assist annotation, but reliability depends on explicit codebooks, validation and audit trails (Gilardi, Alizadeh, & Kubli, 2023; Halterman & Keith, 2024; Ziems et al., 2024).

We address the substantive and methodological questions together. First, we build separate measures of privacy-family prevalence and documented privacy posture. Second, two providers independently process the same frozen source units and codebook. Third, we retain the source-verified union under a one-candidate-one-vote ensemble and show provider-specific sensitivity. Finally, we separate controlled calibration from production validation. This produces a transparent first estimate of institutional associations while making clear which claims remain provisional pending the pre-frozen human audit.

1 Data and method

1.1 Corpus and unitisation

The frozen corpus contains 113 official or authority-attributed CBDC documents, 3,963 pages in total, covering forty-seven national or monetary jurisdictions and two institutional composites. Sources include design reports, consultation papers, technical documents, legislation and official project updates. Each PDF, page image and extracted text unit is identified by a SHA-256 hash. One reference-only document is excluded from extraction. The corpus includes projects at different stages; notably, DCash 2.0 is coded as cancelled after the Eastern Caribbean Central Bank Monetary Council suspended its development in February

2026, while the earlier ECCU design evidence remains in the historical analysis (Eastern Caribbean Central Bank Monetary Council, 2026).

Pages are the base unit. Text is split only when it exceeds the frozen character budget; image renders accompany pages whose text layer is absent or unreliable. All 3,963 units are enumerated in a manifest. The codebook has 35 substantive labels covering privacy, know-your-customer and anti-money-laundering design, architecture, programmability, remuneration, interoperability, access, resilience, legal basis and project learning. Each accepted statement must include an exact source quotation, one primary code, an observation/decision/recommendation status (ODR), privacy direction and relation where applicable, and strength from 1 to 3.

1.2 Two-model extraction and source verification

OpenAI and Claude received byte-identical source packages, prompts, codebooks and output schemas. They were run independently and blinded to each other, to the calibration reference and to the human sample. Provider outputs were archived before parsing. Unsupported spans triggered at most one source-only retry; deterministic finalisation restored whitespace to exact source text, rejected unresolved spans and preserved render-review flags. The final canonical outputs contain 5,675 OpenAI and 5,683 Claude statements. Their union is deduplicated by source-span matching into 6,949 candidates: 4,307 emitted by both providers, 1,315 only by OpenAI and 1,327 only by Claude.

Before production, the exact protocol was tested on 78 calibration pages. OpenAI achieved precision 1.000 and recall 0.625; Claude achieved 0.977 and 0.656; the source-verified union achieved 0.980 and 0.766. Span fidelity was 1.000. The union missed 15 of 64 probability-sample positive paragraphs, and its document-cluster recall interval was 0.633–0.881. These results passed the precision and span gates but failed the frozen recall thresholds. The sealed 40-page reserve was therefore not opened. Production output is analysed as an exploratory, source-verified candidate dataset with sampled human validation, not as a validated automated census.

Tab. 1: Extraction, agreement and validation status

Quantity	OpenAI	Claude	Verified union / design
Production output	5,675 statements / 5,622 candidates	5,683 / 5,634	11,358 statements; 6,949 unique candidates
Calibration precision	1.000	0.977	0.980
Calibration recall (95% CI for union)	0.625	0.656	0.766 [0.633, 0.881]
Calibration span fidelity	1.000	1.000	1.000
Exact code-set agreement	—	—	41.5% among 4,307 shared candidates
Human probability sample	—	—	365/6,949 (5.25%), both authors
Human results	PENDING	PENDING	TO BE FILLED AFTER BLIND LOCK

Source: authors' computation; human fields remain blank until both reviews are locked.

1.3 Ensemble rule and sampled human validation

Every deduplicated candidate contributes total analytic mass 1. A sole provider receives that mass. Where both providers emit the candidate, each receives 0.5; a provider's share is divided equally among its distinct codes, and repeated evidence for one code cannot increase weight. Conflicting categorical labels are retained as fractional votes, never averaged into a synthetic category. Decisions receive ODR weight 1 and proposals 0.5; findings remain in descriptive distributions but not in commitment scores. The ensemble is the primary analysis. Complete OpenAI-only, Claude-only and exact-code-consensus variants are robustness checks.

Human validation uses a deterministic, stratified probability sample selected before coding. Both authors independently review the same 365 candidates (5.2526% of the union), stratified by language and provider origin, plus 36 of 359 units where both models returned no claim. The candidate sample has a worst-case finite-population-corrected 95% margin of 4.99 percentage points. Provider identity and machine labels are hidden. After both files are returned and hash-locked, disagreements alone are adjudicated. Population estimates use inverse inclusion probabilities. This audit estimates candidate validity and code agreement; the dual-empty sample estimates missed-claim yield only in that frame, while controlled retrieval recall remains the calibration estimate (Krippendorff, 2019).

DATA PENDING BEFORE SUBMISSION: insert design-weighted keep rate and 95% CI; Martin–Dominik pre-consensus agreement, Krippendorff's α and Gwet's AC1; provider-to-consensus exact-code agreement; and the dual-empty missed-claim rate/yield.

1.4 Measures and statistical specification

The vocabulary measure is the weighted PRIV-family share among privacy, AML, KYC and technical families. Documented privacy posture is a signed aggregate of source-anchored design choices: direction is positive, negative, conditional or neutral; visibility from the state receives full relation weight and visibility from intermediaries half weight; ODR, strength and ensemble allocation multiply the contribution. Six purpose centres—cash substitution, financial inclusion, payment modernisation, monetary transmission, sovereignty and competition, and state control—combine the frozen code map with lexical signals. Ensemble percentile thresholds calibrate all provider variants on the same scale.

We report pairwise-complete Pearson correlations with V-Dem liberal democracy (Coppedge et al., 2024), central-bank independence (Romelli, 2024), informal-economy size (Elgin, Kose, Ohnsorge, & Yu, 2021), and the Basel AML risk score (Basel Institute on Governance, 2024). Confidence intervals use 10,000 bootstrap draws, p-values use 10,000 permutations, and Holm correction covers the eight primary institution–measure tests.

Associations are descriptive; the small and selected jurisdiction sample does not identify causal effects.

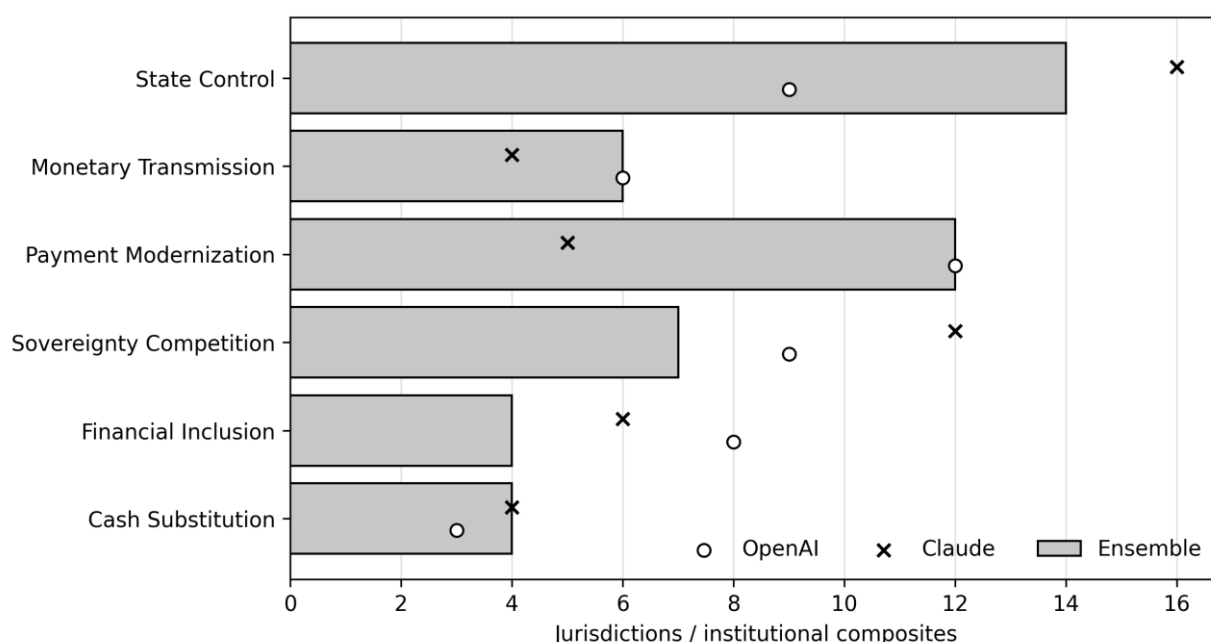
2 Results

2.1 Model overlap and composition

The providers agree on the exact code set for 1,788 of 4,307 shared candidates (41.5%), on the code family for 44.9%, and on ODR status for 62.8%. Single-provider candidates represent 38.0% of the 6,949-candidate union. These are not error rates: a candidate may be valid despite being found by only one model, and two codes may both be defensible. They do show why consensus-only analysis would discard substantial evidence and why silently selecting one provider would embed provider-specific structure.

Figure 1 shows the dominant purpose centre for the 47 entities with decision or proposal evidence. Under the ensemble, state control is dominant for 14, payment modernisation for 12, sovereignty and competition for 7, monetary transmission for 6, and cash substitution and financial inclusion for 4 each. Provider-only classifications vary materially: state-control dominance ranges from 9 under OpenAI to 16 under Claude, while payment modernisation ranges from 5 to 12. The ensemble point estimate is therefore accompanied by the provider markers rather than presented as a uniquely true partition.

Fig. 1: Dominant CBDC purpose by ensemble and provider



Source: authors' computation. Bars are ensemble counts; markers are provider-only sensitivity estimates. Forty-seven entities have decision/proposal evidence.

2.2 Privacy posture and institutional conditions

Privacy vocabulary and documented posture are weakly related. In the ensemble their correlation is $r = 0.219$ ($n = 39$; 95% bootstrap CI $[-0.102, 0.525]$), so privacy-family frequency explains only 4.8% of posture variance. Provider variants remain modest ($r = 0.195$ OpenAI; 0.315 Claude). The distinction is therefore empirically consequential: a document's attention to privacy is not a reliable proxy for the informational boundary its proposals and decisions create.

No institutional correlate survives multiplicity correction (Table 2). Liberal democracy is positively but imprecisely related to posture ($r = 0.152$), not evidence that democracies systematically design more private CBDCs. The largest commitment-side association is informal-economy size ($r = -0.286$), consistent with stronger demand for traceability where unobserved activity is larger, but its interval touches zero and Holm-adjusted p is 0.651. Central-bank independence and Basel risk are near zero on posture. On vocabulary, the largest point estimate is Basel risk ($r = -0.292$), also non-significant.

Tab. 2: Ensemble correlations with privacy vocabulary and posture

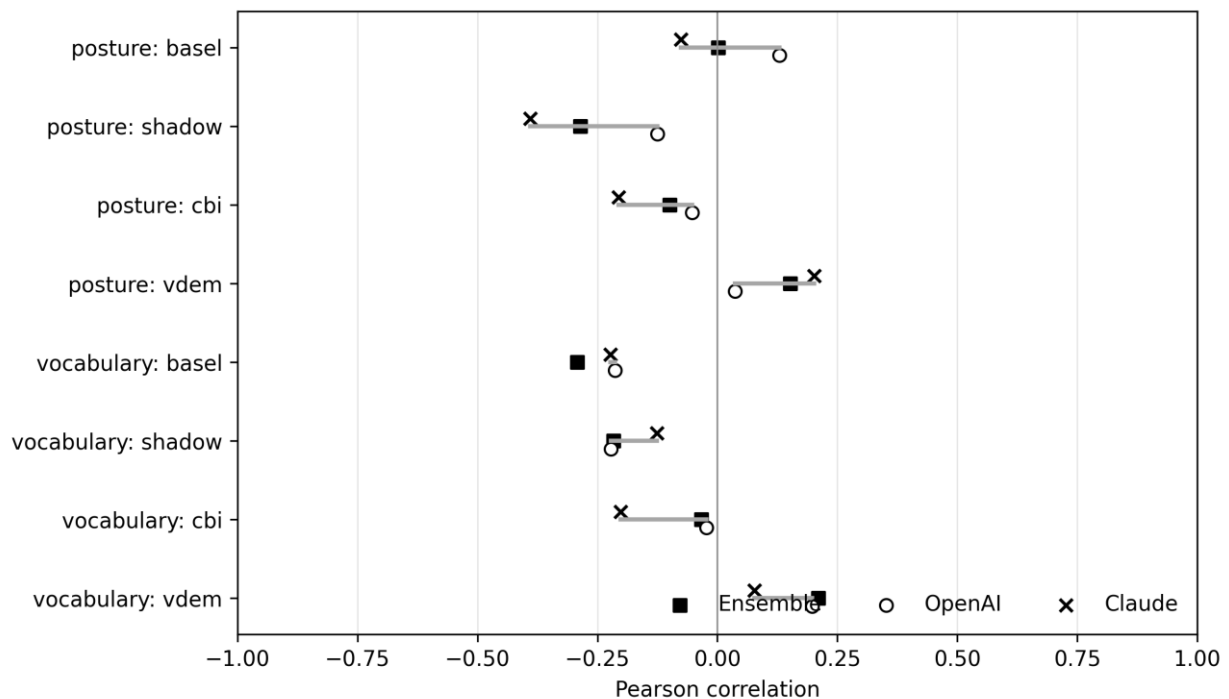
Measure	Condition	r	95% bootstrap CI	Holm p
Vocabulary	Liberal democracy	0.210	$[-0.137, 0.488]$	1.000
Vocabulary	Central-bank independence	-0.033	$[-0.392, 0.336]$	1.000
Vocabulary	Informal economy	-0.216	$[-0.515, 0.150]$	1.000
Vocabulary	Basel AML risk	-0.292	$[-0.582, 0.016]$	0.748
Posture	Liberal democracy	0.152	$[-0.157, 0.417]$	1.000
Posture	Central-bank independence	-0.099	$[-0.561, 0.297]$	1.000
Posture	Informal economy	-0.286	$[-0.532, 0.007]$	0.651
Posture	Basel AML risk	0.002	$[-0.235, 0.255]$	1.000

Source: authors' computation; pairwise $n = 31-38$. Permutation p -values are Holm-corrected over eight primary tests.

2.3 Provider sensitivity

Figure 2 makes model dependence visible. Most institutional correlations remain small under both providers, but some ranges are substantively wide. The posture–informal-economy correlation is -0.125 under OpenAI and -0.390 under Claude; the ensemble lies between them at -0.286 . The vocabulary–democracy estimate ranges from 0.077 to 0.198, while the consensus-only subset reverses sign (-0.203). Consensus is thus a selected high-overlap subset, not a neutral adjudicator. The primary ensemble preserves recall; provider and consensus variants reveal how much interpretation depends on model selection.

Fig. 2: Provider sensitivity of institutional correlations



Source: authors' computation. Squares are ensemble estimates; circles and crosses are provider-only estimates. Grey lines show the provider range.

3 Discussion

Substantively, the current evidence does not support a strong regime account of CBDC privacy. The democracy coefficient is small, uncertain and sensitive to whether privacy is measured as vocabulary or posture. Informal-economy size remains the most plausible structural correlate, but the corrected evidence is insufficient for a finding. The safer conclusion is negative but useful: institutional narratives built from a few visible projects are not recovered robustly in this larger document corpus.

Methodologically, the exercise shows why multilingual LLM extraction should be treated as measurement, not clerical automation. Source verification nearly eliminates unsupported evidence, but it does not guarantee that all eligible evidence was found or correctly classified. The calibration union combined high precision with recall 0.766 and failed the predeclared gate. Independent models improved coverage, yet exact-code agreement remained 41.5% among shared candidates. Reporting only one model, or only consensus cases, would hide this uncertainty. The equal-vote union makes it explicit and bounded.

Four limitations matter. First, the corpus is selected by document availability and central-bank transparency; early-stage and low-capacity jurisdictions are underrepresented. Second, official publications are strategic communications and may not predict final implementation. Third, source-verified production candidates remain machine coded until the sampled human audit is completed; this draft must not be submitted with the pending fields unresolved. Fourth,

the dual-empty audit covers only units where neither model extracted a claim, not misses inside otherwise non-empty units. These limits preclude a full-production recall estimate and strong causal claims.

Conclusion

CBDC privacy cannot be inferred reliably from how often official documents use privacy language. In the v10.2e ensemble, vocabulary and documented posture share less than 5% of variance, and none of four political or economic conditions survives family-wise correction. The informal economy produces the largest posture-side association, but it remains a hypothesis for larger and better-balanced samples. The result is not that institutions never matter; it is that the present public record does not identify a stable macro determinant.

The larger contribution is a reproducible boundary around LLM-assisted qualitative research. Two independent providers, frozen source packages, exact-span verification, archived raw outputs, an explicit ensemble rule and a probability-sampled human audit turn model output into inspectable evidence. They also reveal failure: high precision did not compensate for inadequate calibrated recall. The final paper will replace the shaded placeholders with locked human-validation estimates; the substantive tables will retain the full union and disclose provider sensitivity rather than erase disagreement.

Data and code availability

The frozen code, manifests, provider archives and derived outputs will be released at <https://github.com/BokMartin/cbdc-msed-v10>. A browsable project website is prepared for deployment at <https://bokmartin.github.io/cbdc-msed-v10/> [TO BE ACTIVATED AND VERIFIED BEFORE SUBMISSION].

References

- Agur, I., Ari, A., & Dell’Ariccia, G. (2022). Designing central bank digital currencies. *Journal of Monetary Economics*, 125, 62–79. <https://doi.org/10.1016/j.jmoneco.2021.05.002>
- Auer, R., & Böhme, R. (2020). The technology of retail central bank digital currency. *BIS Quarterly Review*, March, 85–100.
- Auer, R., Cornelli, G., & Frost, J. (2023). Rise of the central bank digital currencies. *International Journal of Central Banking*, 19(4), 185–214.
- Basel Institute on Governance. (2024). Basel AML Index 2024: 13th public edition—Ranking money laundering and terrorist financing risks around the world.
- Coppedge, M., Gerring, J., Knutsen, C. H., Lindberg, S. I., Teorell, J., et al. (2024). V-Dem Country–Year Dataset v14. Varieties of Democracy Project.
- Eastern Caribbean Central Bank Monetary Council. (2026, February 13). Communiqué of the 112th meeting of the Monetary Council of the Eastern Caribbean Central Bank. <https://www.eccb-centralbank.org/news/communique-of-the-112th-meeting-of-the-eccb-monetary-council>
- Elgin, C., Kose, M. A., Ohnsorge, F., & Yu, S. (2021). Understanding informality (CEPR Discussion Paper No. 16497). Centre for Economic Policy Research.

- Garratt, R. J., & van Oordt, M. R. C. (2021). Privacy as a public good: A case for electronic cash. *Journal of Political Economy*, 129(7), 2157–2180. <https://doi.org/10.1086/714133>
- Gilardi, F., Alizadeh, M., & Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proceedings of the National Academy of Sciences*, 120(30), e2305016120. <https://doi.org/10.1073/pnas.2305016120>
- Halterman, A., & Keith, K. A. (2024). Codebook LLMs: Adapting political science codebooks for LLM use and adapting LLMs to follow codebooks. *arXiv:2407.10747*.
- Kahn, C. M., McAndrews, J., & Roberds, W. (2005). Money is privacy. *International Economic Review*, 46(2), 377–399. <https://doi.org/10.1111/j.1468-2354.2005.00323.x>
- Kosse, A., & Mattei, I. (2022). Gaining momentum: Results of the 2021 BIS survey on central bank digital currencies (BIS Papers No. 125). Bank for International Settlements.
- Krippendorff, K. (2019). *Content analysis: An introduction to its methodology* (4th ed.). SAGE Publications.
- Romelli, D. (2024). Trends in central bank independence: A de-jure perspective (BAFFI CAREFIN Working Paper No. 217). Bocconi University.
- Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., & Yang, D. (2024). Can large language models transform computational social science? *Computational Linguistics*, 50(1), 237–291. https://doi.org/10.1162/coli_a_00502

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